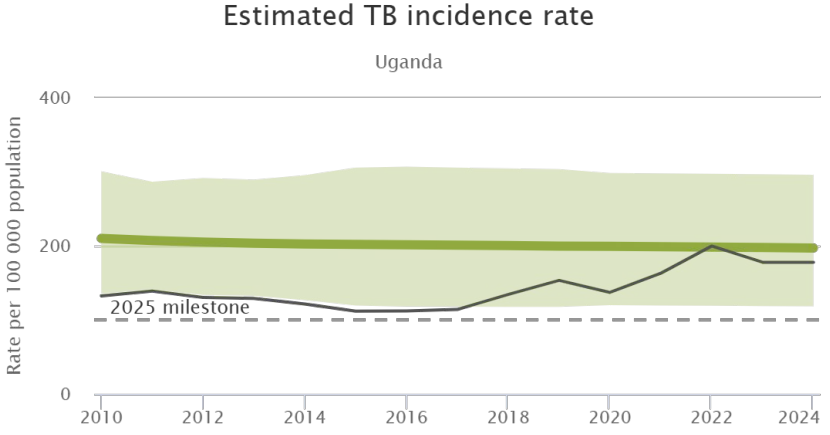
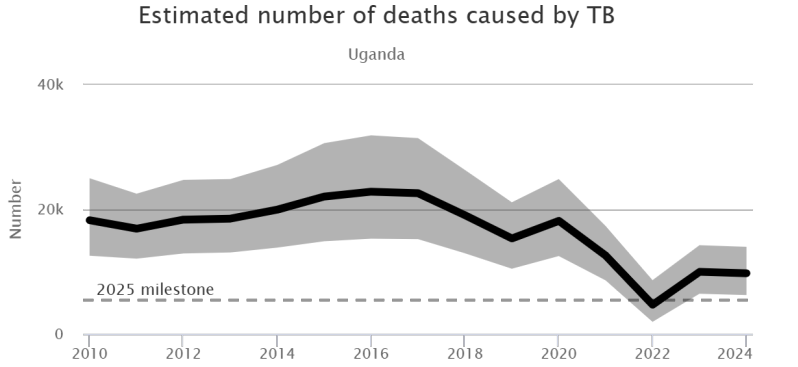
# Bioreactor

By Group CSEEE-21

# Context:

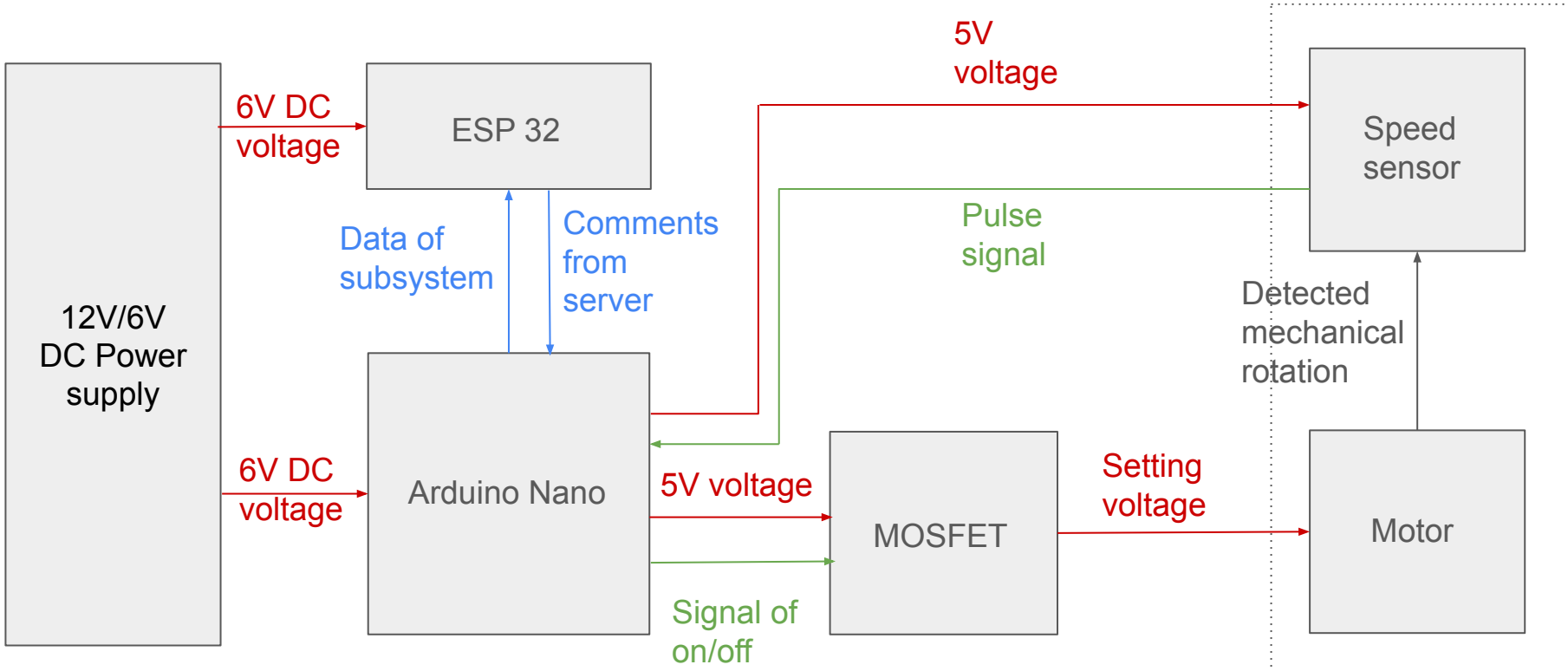
- Tuberculosis(TB) is a bacterial infection usually affecting the lungs
- Most common treatment of TB is through special antibiotics, such as rifampicin or isoniazid etc.
- Only effective if the TB they have contracted is not Multi-drug resistant(MDR).
- “Only about 2 in 5 people with multidrug-resistant TB accessed treatment in 2024.”, according to the WHO.
- Prevention may also be an option
- A vaccine trains the immune system to attack the actual bacterium so isn't affected by the various strains
- Often cheaper as it is a one-time dose compared to the months of treatment on antibiotic



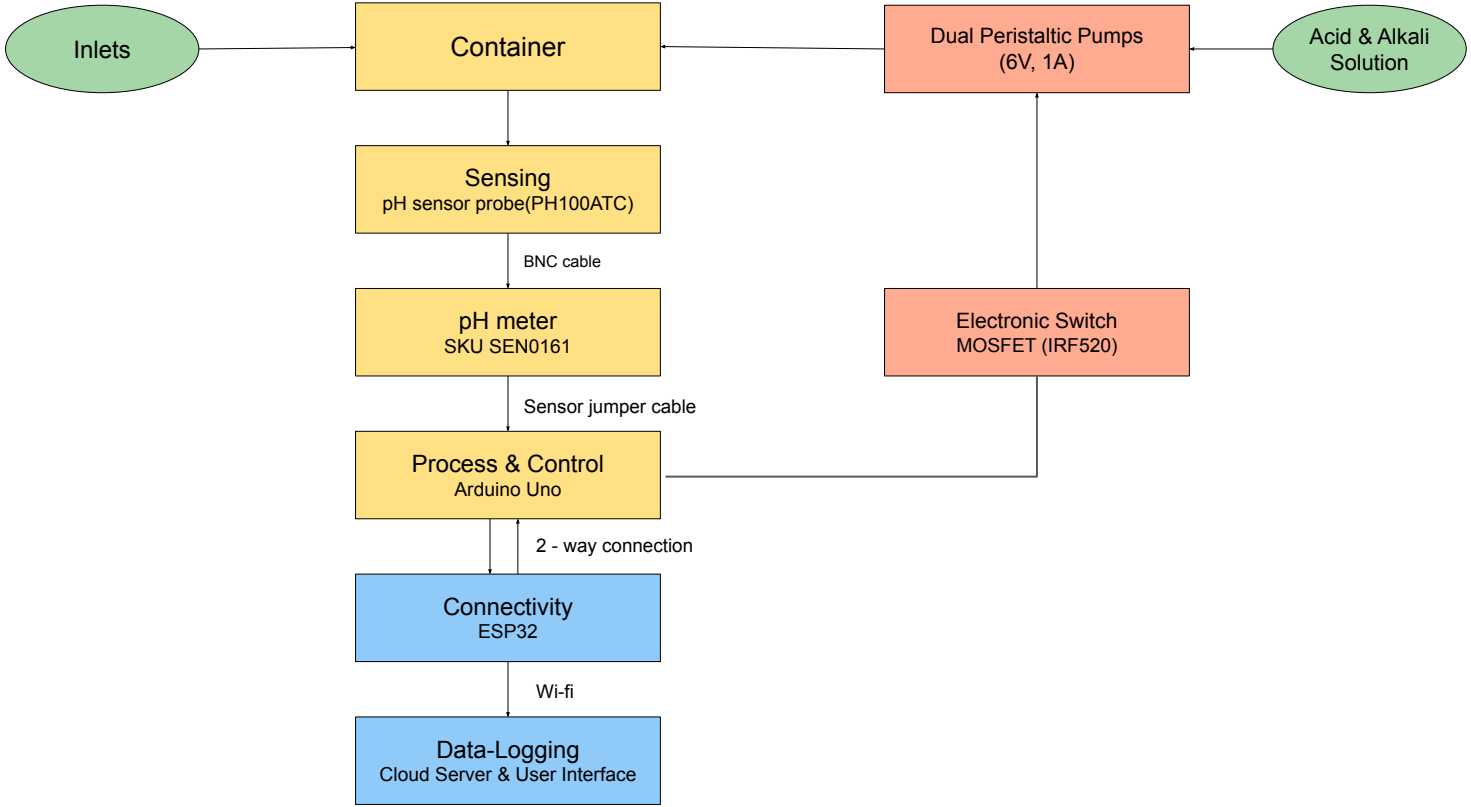




Source:https://worldhealthorg.shinyapps.io/tb_profiles/?_inputs_&tab=%22charts%22&lan=%22EN%22&iso3=%22UGA%22&entity_type=%22country%22

# Stirring subsystem

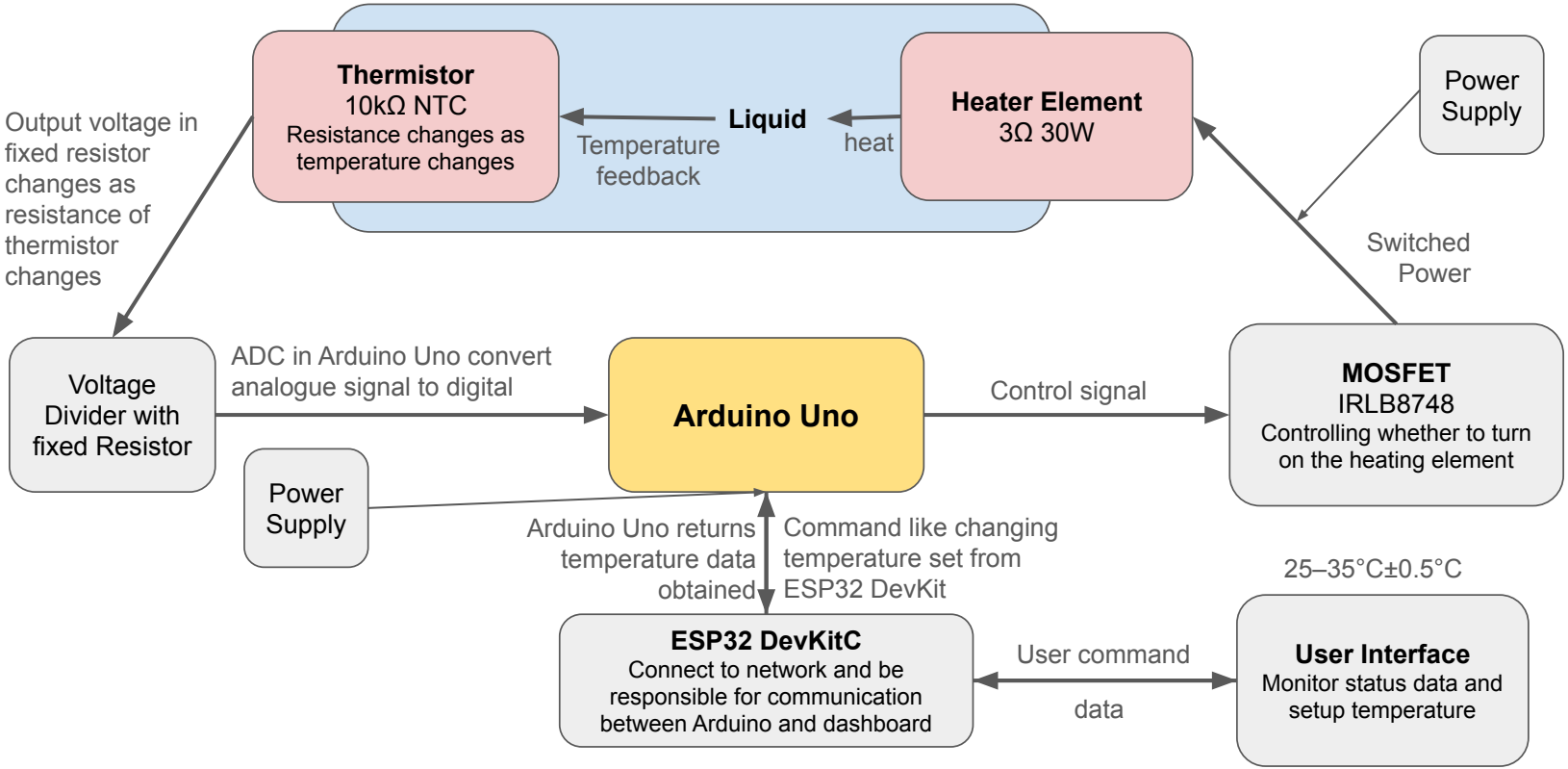


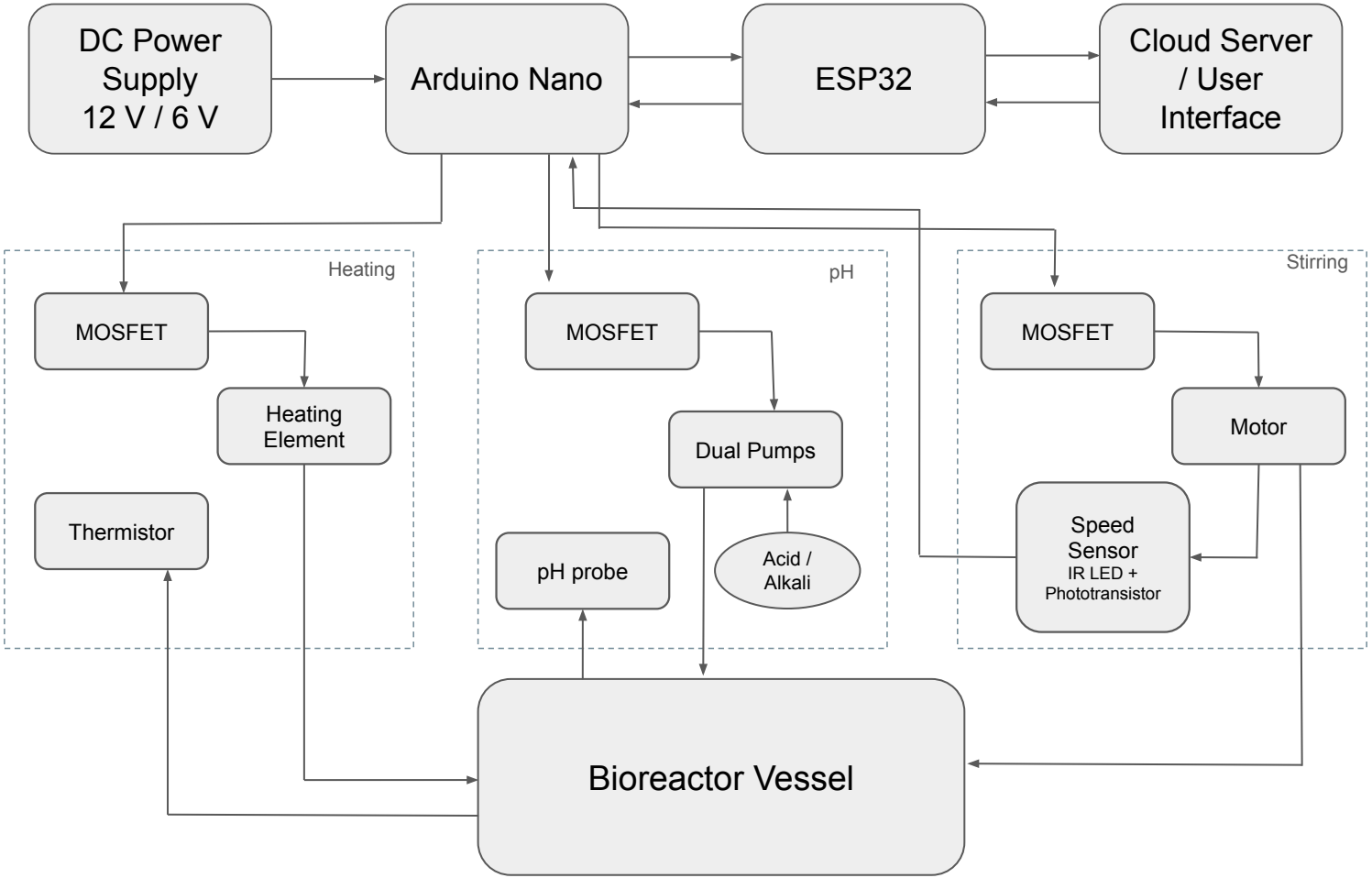
# pH subsystem




By: abie

# Heating subsystem




DC Power Supply 12 V / 6 V
Arduino Nano
ESP32
Cloud Server / User Interface
Heating
MOSFET
Heating Element
Thermistor
pH
MOSFET
Dual Pumps
pH probe
Acid / Alkali
Stirring
MOSFET
Motor
Speed Sensor
IR LED + Phototransistor
Bioreactor Vessel

# Connectivity and User Interface

Alternative proposal: direct HTTP over a local Wi-Fi connection

**AT THE REACTOR** | **ON THE OPERATOR LAPTOP**

**Control board**
Sensors + actuators
Local feedback loops

UART ↔

**ESP32 gateway**
Wi-Fi access point
HTTP API + UART bridge

Wi-Fi / HTTP
JSON requests ↔

**Laptop backend**
Polls state + sends targets
Hosts UI + CSV logs

Local web ↔

**Browser**
Dashboard
Operator inputs

Laptop joins the ESP32 network. Backend polls readings and sends target requests.

**PROPOSED SCREEN / example values**

**Temperature**
**29.8 °C**
Confirmed: 30.0 °C
New target | Apply

25–35 °C

**Stirring speed**
**998 RPM**
Confirmed: 1000 RPM
New target | Apply

500–1500 RPM

**pH**
**5.02**
Confirmed: 5.00
New target | Apply

pH 3–7

**OPERATOR ACTIONS**

Monitor values and trends.

Apply validated setpoints.

Wait for controller confirmation.

Export timestamped records.

**On link loss: mark readings stale.**
Local loops follow the agreed policy.

PROPOSED TESTS 1 Hz reporting | 95% of confirmations ≤2 s | stale warning after 5 s

Local access only: confirm this scope and access-point use with staff. Internet access is not provided by this arrangement.

# Data Analysis

- Using hysteresis two thresholds
- Calculate median and MAD for modified z-score
- Thresholds decided using anomalous data

